Check for updates

Gravitational waves confirm predicted black-hole mass gap and probe stellar nuclear physics

The mass spectrum of binary black-hole mergers has been expected to show a 'mass gap' above 45 solar masses, consistent with the physics of pair-instability supernovae. An extensive catalogue of gravitational-wave detections reveals a high-spin population above this threshold that probably results from repeated black-hole mergers in dense star clusters, and fits with the existence of the mass gap. The inferred boundary also constrains the carbon-to-oxygen fusion rate during helium burning in massive stars.

This is a summary of:
Antonini, F. et al. Gravitational-wave constraints on the pair-instability mass gap and nuclear burning in massive stars. *Nat. Astron.* https://doi.org/10.1038/s41550-026-02847-0 (2026).

Published online: 01 June 2026

The problem

A massive star may undergo pair instability: gamma rays in the core produce electron–positron pairs, reducing radiation pressure and triggering partial collapse. This can lead either to violent pulses that eject mass before the star finally collapses, or to a full explosion that leaves no remnant at all. As a result, pair instability suppresses black-hole formation over a range of masses, giving rise to the expected 'mass gap' in the black-hole mass spectrum. Yet gravitational-wave observations have repeatedly identified black holes in this forbidden range, raising questions about whether the gap actually exists and how these black holes have formed. Resolving this tension is central to understanding both stellar evolution and the environments in which black holes grow.

At the same time, the location of the mass gap depends sensitively on the nuclear reactions inside the star during helium burning – linking gravitational-wave observations to fundamental nuclear physics. Using the latest catalogue of recorded gravitational waves, we set out to determine whether distinct black-hole populations emerge across this boundary and to assess what this reveals about the origin of the gravitational-wave sources and the physics of stellar interiors.

The discovery

We have analysed the fourth LIGO–Virgo–KAGRA gravitational-wave transient catalogue[1] (GWTC-4), which contains more than double the number of observed binary black-hole mergers compared with the previous catalogue, enabling population studies with unprecedented precision[2]. Using hierarchical inference, we modelled the distribution of the effective spin parameter (an effective combination of the two black hole spins and the best measured spin parameter from the data) as a function of primary mass, combining Gaussian-process and parametric approaches to enable flexible, data-driven inferences of structure in the black-hole mass and spin distributions. This framework was designed to identify transitions in the population without imposing strong prior assumptions, and to test whether distinct formation channels emerge at high masses. By linking spin and mass distributions, our approach provided a direct way to distinguish 'first-generation' black holes (formed from the merger of only two black holes) from those formed through repeated mergers, while also enabling constraints on the physical processes that set the pair-instability mass scale.

We found clear evidence for a transition at approximately 45 solar masses (Fig. 1), marking the lower edge of the pair-instability mass gap at a value that has long been theoretically predicted. Below this threshold, black holes have low, preferentially aligned spins consistent with the merger of black holes formed directly from stellar collapse; above the threshold, the population shows broad, isotropic spins indicative of repeated black-hole mergers in dense star clusters[3]. The data reveal a sharp drop in the merger rate near this boundary, and a high-mass population that extends into the gap. Interpreting the transition as the onset of pair instability, we also inferred the rate of helium capture by carbon that forms oxygen in massive stars. These results not only provide strong evidence for two distinct black-hole populations but also establish a direct observational link between gravitational-wave astronomy and stellar nuclear physics.

The implications

Our results show that gravitational-wave observations can probe not only black-hole formation, but also the internal physics of massive stars. By linking the black-hole mass spectrum to helium-burning processes, we have provided an independent astrophysical constraint on a key nuclear reaction that shapes stellar evolution, supernova outcomes and chemical enrichment[4,5]. More broadly, identifying a hierarchical merger population highlights dense stellar clusters as crucial sites of black-hole growth across cosmic time.

Our interpretation relies on the assumption that black holes above the inferred transition mass of ~45 solar masses are predominantly products of hierarchical mergers. Alternative stellar-evolution pathways could, in principle, contribute to this population, and uncertainties in spin measurements and selection effects might influence the inferred distributions. In addition, translating the mass-gap location into a nuclear reaction rate depends on stellar models, which carry their own systematic uncertainties.

Future observations with larger gravitational-wave catalogues will sharpen constraints on the mass gap and spin distributions, enabling more precise tests of the channels of black-hole formation. Extending this framework to include mass ratios and redshift evolution, and comparing with detailed cluster simulations, will help disentangle the relative contributions of different astrophysical environments.

Fabio Antonini & Isobel Romero-Shaw
Cardiff University, Cardiff, Wales, UK.

EXPERT OPINION

"The proposed formation scenario based on the pair-instability supernova mechanism offers a compelling explanation for trends observed in LIGO–Virgo–KAGRA detections, effectively synthesizing insights gained over roughly a decade of gravitational-wave observations." **Aleksandra Olejak, Max Planck Society, Munich, Germany.**

FIGURE

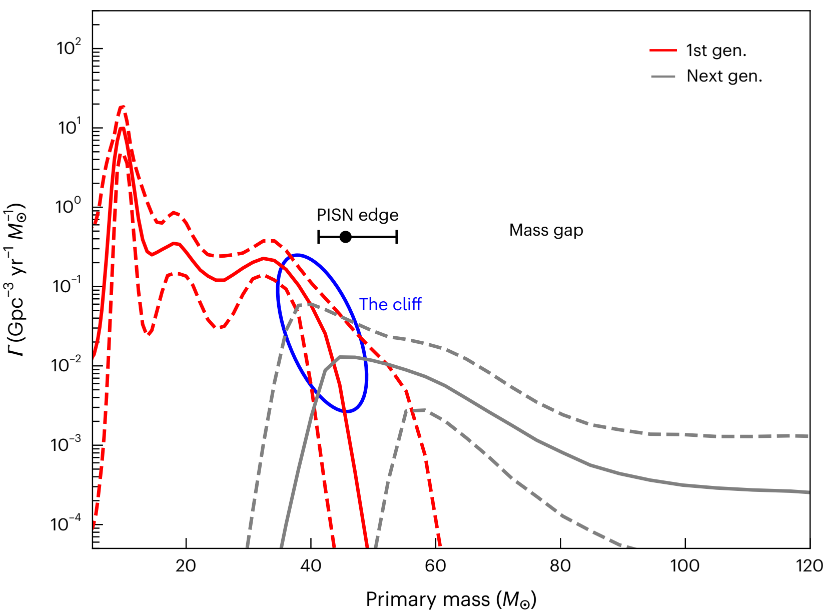

Fig. 1 | Black-hole merger rate across the pair-instability mass scale. The merger-rate density (Γ) is shown as a function of primary black-hole mass at redshift $z = 0.2$, separated into low-spin (red) and high-spin (black) populations. Solid lines show median rates; dashed lines indicate the 10th and 90th percentiles. The transition mass is marked by the black point with error bars (90% confidence level) and corresponds to the onset of a high-spin, isotropic population. The sharp drop, or 'cliff', and subsequent high-mass component are consistent with a pair-instability mass gap that is populated by black-hole mergers occurring in dense star clusters. PISN, pair-instability supernovae.

BEHIND THE PAPER

The project grew out of a simple tension that would not go away: theory predicts that pair instability should prevent black holes from forming in a certain mass range, yet gravitational-wave catalogues keep recording candidates in exactly that region. Our earlier work on the previous catalogue (GWTC-3) hinted that something important was happening close to 45 solar masses, but the sample was too small to be definitive. The real turning point came with GWTC-4 — once the catalogue had more than doubled in size, the transition in spin properties and mass distribution became difficult to ignore.

What made the collaboration especially productive was that it brought together complementary expertise in gravitational-wave population inference, stellar dynamics and stellar evolution. The hardest part was convincing ourselves that the feature was truly driven by the data and not by modelling choices. Realizing that the same boundary could also inform helium-burning physics was the genuine 'eureka' moment. **F.A.**

REFERENCES

1. Abac, A. G. et al. GWTC-4.0: Updating the gravitational-wave transient catalog with observations from the first part of the fourth LIGO–Virgo–KAGRA observing run. Preprint at https://arxiv.org/abs/2508.18082 (2025).
This preprint presents the catalogue of gravitational wave sources discovered in the LIGO–Virgo–KAGRA fourth observing run.
2. Abac, A. G. et al. GWTC-4.0: Population properties of merging compact binaries. Preprint at https://arxiv.org/abs/2508.18083 (2025).
This preprint explores the population properties of the gravitational-wave sources discovered up to the fourth observing run.
3. Antonini, F., Romero-Shaw, I. M. & Callister, T. Star cluster population of high-mass black hole mergers in gravitational-wave data. *Phys. Rev. Lett.* **134**, 011401 (2025).
This paper provides evidence for the effective-spin transition at high black-hole masses in the GWTC-3 catalogue.
4. Woosley, S. E. Pulsational pair-instability supernovae. *Astrophys. J.* **836**, 244 (2017).
This paper presents a study example of the physics of pair-instability in massive stars.
5. deBoer, R. J. et al. The $^{12}C(\alpha,\gamma)^{16}O$ reaction and its implications for stellar helium burning. *Rev. Mod. Phys.* **89**, 035007 (2017).
This review article covers experimental and theoretical constraints on the key helium-burning reaction that converts carbon into oxygen.

FROM THE EDITOR

"Leveraging the new gravitational-wave catalogue to not only identify the pair-instability mass gap, but also to separate out two binary black-hole populations and set constraints on nuclear burning properties, makes this paper stand out." **Lindsay Oldham, Associate Editor, *Nature Astronomy*.**